Week 9: K-Nearest Neighbors (k-NN)

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Supervised Learning: Overview

- **Regression**:
  - Linear Regression, Model representation: $\theta^T x$
  - Cost Function: Mean Squared Error (MSE)
  - Evaluation Metric: MSE
- **Classification**:
  - Logistic Regression, Model representation: $\sigma(\theta^T x)$
  - Cost Function: Negative Log-Likelihood Loss (NLLL)
  - Evaluation Metric: Confusion Matrix

k-NN

Assumption: Similar inputs should have similar outputs.

- Model representation: ?
- Cost function: ?
- Evaluation Metric:
  - Confusion Matrix (for classification)
  - MSE (for regression)

Properties:

- Non-parametric models
- Lazy learning / instance-based method

k-NN Algorithm

1. Find distances between the testing sample and all training samples using Euclidean method.

$$d(x_i, x_j) = \sqrt{\sum_{k=1}^{d}(x_{i,k} - x_{j,k})^2}, \text{ where } i \neq j$$

2. Sort the distances from minimum to maximum
3. Pick k nearest neighbors
4. Use majority vote for classification or average for regression for that testing sample.

k-NN for Regression and Classification

Sample Data:

Example dataset	x_1	x_2	y_1	y_2
s1	1	10	1000	yes
s2	2	7	1200	yes
s3	3	5	1100	no
s4	3	2	2000	yes
s5	**2**	**9**		

Using the k-NN to find $y1$ and $y2$ of $s5$ for

k $= 1$,
k $= 2$,
k $= 3$,
k $= 4$,

k-NN for Categorical Data (1/2)

Example dataset

Sample	Food	Chat	Fast	Price	Bar	Tip
s1	great	yes	yes	normal	no	yes
s2	great	no	yes	normal	no	no
s3	mediocre	yes	no	high	no	no
s4	great	yes	yes	normal	yes	yes
s5	great	no	no	normal	no	

Distance: **Hamming method** Match = 0, Mismatch = 1

Definition

The Hamming distance between two equal-length strings of symbols is the number of positions at which the corresponding symbols are different.

Examples

The symbols may be letters, bits, or decimal digits, among other possibilities.
For example, the Hamming distance between:

- "**karolin**" and "**kathrin**" is 3.
- "**karolin**" and "**kerstin**" is 3.
- "**kathrin**" and "**kerstin**" is 4.
- **0000** and **1111** is 4.
- **2173896** and **2233796** is 3.

k-NN for Categorical Data (2/2)

Example dataset

Sample	Food	Chat	Fast	Price	Bar	Tip
s1	great	yes	yes	normal	no	yes
s2	great	no	yes	normal	no	yes
s3	mediocre	yes	no	high	no	no
s4	great	yes	yes	normal	yes	yes
s5	great	no	no	high	no	

Distance: Hamming method Match $= 0$, Mismatch $= 1$

Define $k = 2$, then compute:

- $H(s5, s1)$
- $H(s5, s2)$
- $H(s5, s3)$
- $H(s5, s4)$

Tuning the Hyperparameter k

How to choose k?

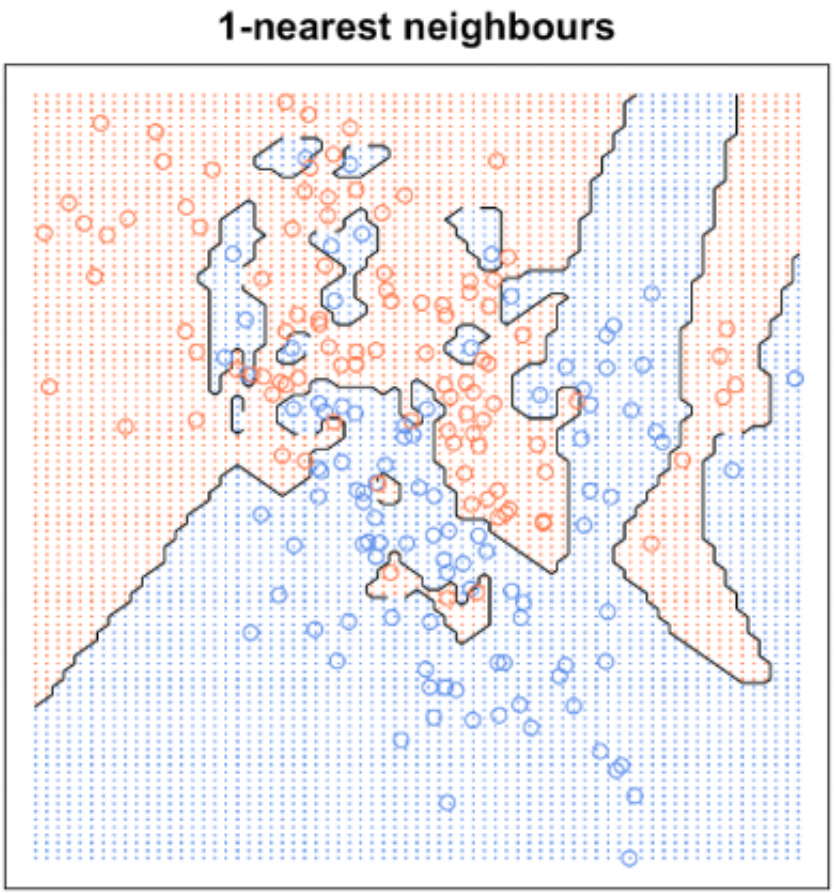
- Use odd k values to avoid ties: $k = 1, 3, 5, ..., \sqrt{N}$, where N is the size of dataset.
- Perform cross-validation using, e.g., `GridSearchCV`

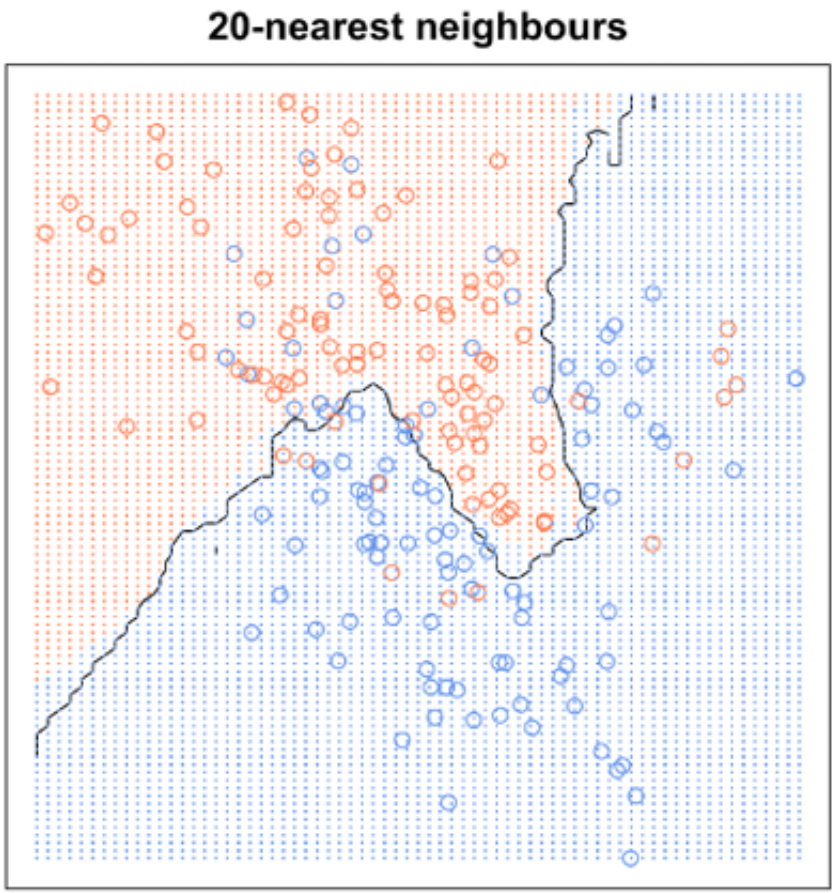
Effect of k:

- k too small: overfitting
- k too large: underfitting

Decision Boundaries

Decision boundary





Ref: https://kevinzakka.github.io/2016/07/13/k-nearest-neighbor/

k-NN issues 1

Issue 1: Scaling Disparity

- Features with large magnitude dominate distance calculations.
- Example: Population = 120,000,000 VS Age = 60

Solution: Feature Scaling

- **z-score normalization**: Maps to range $(-3, 3)$
- **Min-max scaling**: Maps to range $[0, 1]$

k-NN issues (Example of issue 1)

id	X_1	X_2	Y
s_1	4.5	10000	yes
s_2	0.1	3000	no
s_3	0.2	3200	yes
s_4	4.5	3000	

Compare the results to find s_4's class (for k=1) between before and after applying min-max scaling.

k-NN issues (2-4)

Issue 2: Noisy Data

- Solution: Remove outlier data
- Methods: Robust scaling ((x_i - median)/IQR)
- Ref: `https://en.wikipedia.org/wiki/Interquartile_range`

Issue 3: Slow Testing Time for Big Data

- Solution: Use KD-Tree, Ball-Tree, etc
- Ref: `https://www.youtube.com/watch?v=BK5x7IUTIyU`

Issue 4: Storage Requirements

- Solution: Product quantization (PQ)
- Ref: `https://docs.nvidia.com/cuvs/user-guide/api-guides/preprocessing-guide/product-quantization`

k-NN issues (2-4)

Issue 5: Curse of Dimensionality

- As dimensions increase, the volume of the space grows exponentially. Consequently, data becomes sparse, requiring significantly more data points to fill the space and ensure proper model generalization.
- Solution (Dimensionality reduction)
  - PCA
  - LDA
  - t-SNE
  - Autoencoders

Application

- Im2gps: https://graphics.cs.cmu.edu/projects/im2gps/im2gps.pdf
- Data imputation: https://www.sciencedirect.com/science/article/pii/S0164121212001586?casa_token=EQPFMncizwsAAAAA:wF2cst5NhtlUAUHmVEqkg2c8JskyGTFywEUtfUaqSRp6Qi7vux04wG3YYI
- Recommendation system: https://www.sciencedirect.com/science/article/pii/S221083271400026X

Reference

- Bhatia,N.(2010). Survey of nearest neighbor techniques. arXiv preprint arXiv:1007.0085.